IN SILICO ORGANOID TWINS: AUTOMATED BIOINFORMATICS SIMULATION FOR PATIENT-SPECIFIC DRUG SCREENING

Brief Description

Category: Biomedical Engineering and Digital Health Technologies
Project Type: Engineering / Business Domain
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Relevance

The global oncology drug development pipeline faces a critical crisis: over 90% of cancer drug candidates fail in clinical trials, costing an estimated USD 1–5 million per compound annually in false-positive screening alone. Current preclinical models are the root cause. Two-dimensional cell cultures achieve only ~50% sensitivity in predicting clinical responses. Patient-derived organoids (PDOs) improve this to 80–90%, yet remain inaccessible at scale — requiring 4–6 weeks per test cycle, costs exceeding USD 10,000 per panel, and producing fewer than 10,000 organoids per batch with 20–50% variability. A computational alternative would eliminate these bottlenecks entirely.

Objective

To develop, validate, and deploy an automated multi-scale bioinformatics pipeline — In Silico Organoid Twins — that computationally replicates patient-specific organoid drug responses, enabling high-throughput personalised oncology drug screening at 1% of the cost and in 1% of the time of current experimental workflows.

Methods and Technologies

The platform integrates five computational layers: (1) U-Net convolutional neural networks trained on annotated histological biopsy images to reconstruct 3D organoid geometry; (2) single-cell RNA sequencing (scRNA-seq) data integrated into agent-based models via the PhysiCell framework, simulating proliferation, apoptosis, and intercellular signalling across large heterogeneous cell populations; (3) QSAR and PK/PD drug modeling using RDKit and OpenEye to predict molecular properties and pharmacokinetics; (4) PDE solvers (FEniCS) simulating drug diffusion and metabolic consumption within the virtual organoid volume; and (5) GPU-accelerated inference via PyTorch enabling parallel screening of 10^5 virtual compounds per hour. Calibration is performed against 1,000+ real PDO profiles from DepMap and GDSC using Bayesian optimisation (Optuna). The platform is delivered as a Streamlit web application with FHIR-compliant EHR integration, containerised via Docker and Kubernetes for HPC deployment.

Main Results and Implementation Prospects

Projected outcomes include: a 95% reduction in per-test cost (USD 100 vs. USD 10,000); turnaround compressed from 4–6 weeks to under 4 hours; and target concordance of Pearson $r \geq 0.95$ with experimental PDO drug response data. The platform reduces pharmaceutical drug attrition by a projected 30%, saving billions annually across the industry. Local HPC deployment capability enables sovereign pharmaceutical development in resource-limited or sanction-affected settings without dependence on imported biological reagents. Primary customers include pharmaceutical companies, contract research organisations, and national health agencies.

Expected Conclusions and Impact on Medical Science and Practice

This project establishes a scientifically rigorous, commercially viable computational equivalent to physical organoid drug testing. Its successful deployment will: (1) democratise personalised oncology drug screening beyond elite research institutions; (2) accelerate preclinical development cycles and reduce late-stage clinical trial failures; (3) provide oncologists with patient-specific drug sensitivity profiles within hours, enabling evidence-based treatment selection; and (4) contribute a validated simulation engine, annotated training dataset, and benchmarking framework as foundational resources for the emerging field of computational organoid pharmacology. Future development pathways include extension to non-oncology indications, immunological microenvironment simulation, and regulatory pathway development for clinical decision-support approval.